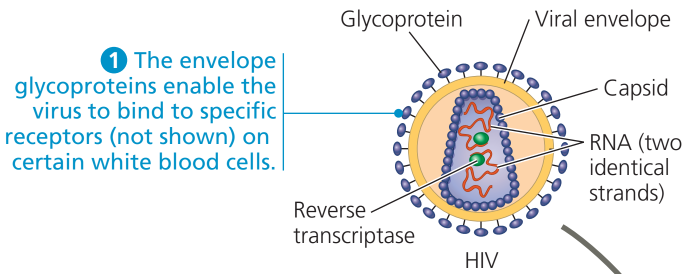
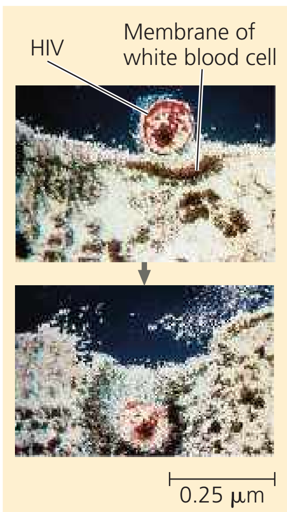
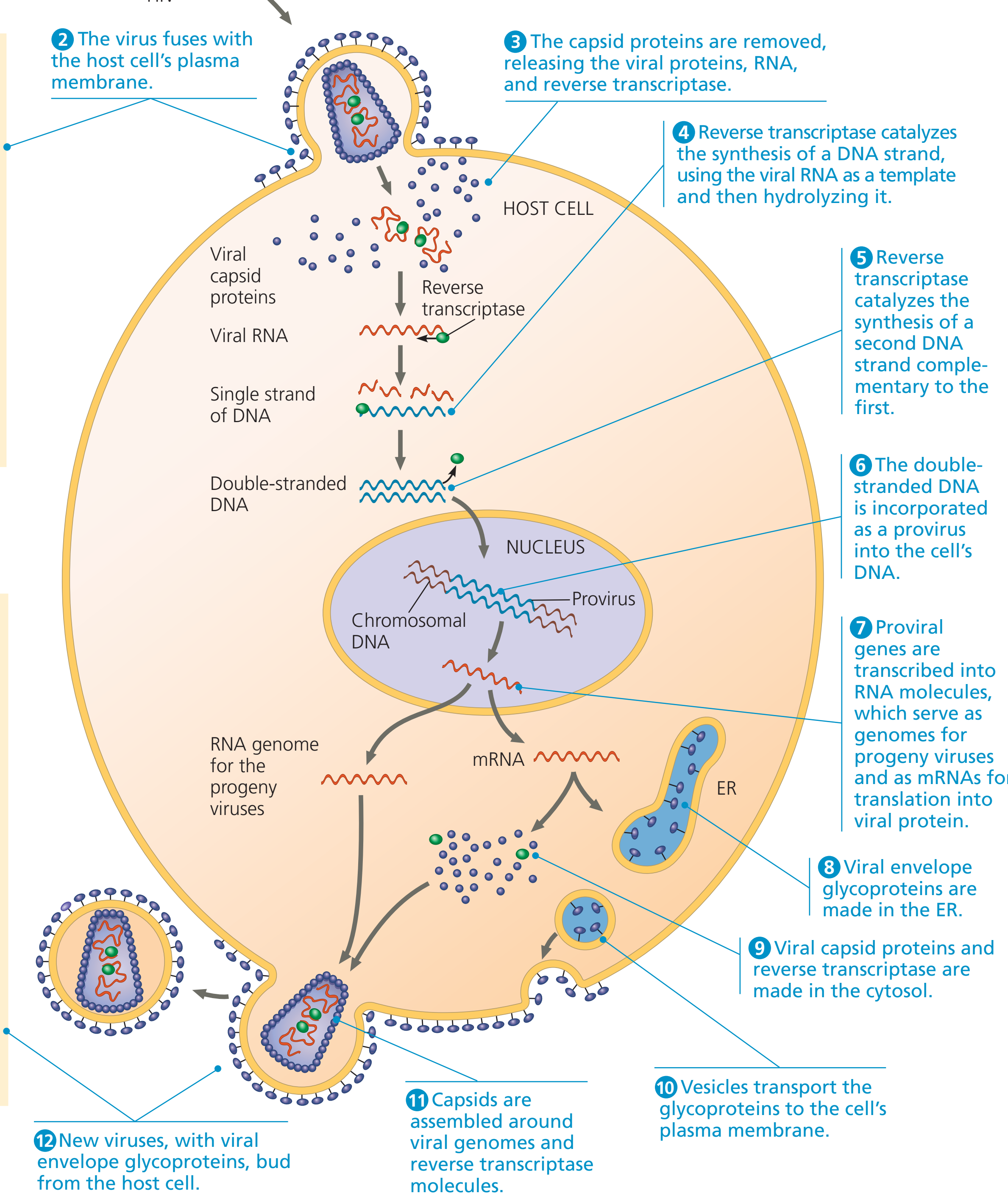
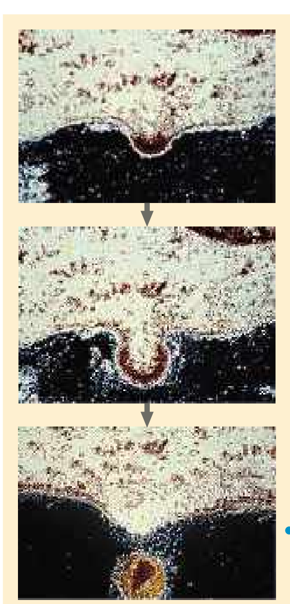
▼ **Figure 19.9 The replicative cycle of HIV, the retrovirus that causes AIDS.** The photos on the left (artificially colored TEMs) show HIV entering and leaving a human white blood cell. See Figure 7.8 for the cell-surface proteins that act as receptors for HIV. Note in step 6 that DNA synthesized from the viral RNA genome is integrated as a provirus into the host cell chromosomal DNA, a characteristic unique to retroviruses.





HIV entering a cell





New HIV leaving a cell

MAKE CONECTIONS *Describe what is known about binding of HIV to immune system cells. (See Figure 7.8.) How was this discovered?*

Mastering Biology **HHMI Animation: HIV Replicative Cycle**
Animation: Retrovirus (HIV) Replicative Cycle